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ORIGINAL PAPER

Where the Action is in Crime? An Examination of Variability of Crime Across Different Spatial Units in The Hague, 2001–2009

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Abstract
Objectives To identify how much of the variability of crime in a city can be attributed to micro (street segment), meso (neighborhood), and macro (district) levels of geography. We define the extent to which different levels of geography are important in understanding the crime problem within cities and how those relationships change over time.
Methods Data are police recorded crime events for the period 2001–2009. More than 400,000 crime events are geocoded to about 15,000 street segments, nested within 114 neighborhoods, in turn nested within 44 districts. Lorenz curves and Gini coefficients are used to describe the crime concentration at the three spatial levels. Linear mixed models with random slopes of time are used to estimate the variance attributed to each level.
Results About 58–69 % of the variability of crime can be attributed to street segments, with most of the remaining variability at the district level. Our findings suggest that micro geographic units are key to understanding the crime problem and that the neighborhood does not add significantly beyond what is learned at the micro and macro levels. While the total number of crime events declines over time, the importance of street segments increases over time.
Conclusions Our findings suggest that micro geographic units are key to understanding the variability of crime within cities—despite the fact that they have received little criminological focus so far. Moreover, our results raise a strong challenge to recent focus on such meso geographic units as census block groups.

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Introduction

Over the last few decades, criminologists have begun to recognize the importance of micro geographic units in the study and prevention of crime (Bursik and Grasmick 1993; Eck and Weisburd 1995; Hipp 2010; Sherman and Weisburd 1995; Tita and Radil 2010; Weisburd et al. 2012). A key reason for this has been the startling evidence surrounding the concentration of crime at a small number of places in a city. Beginning in the late 1980s, a series of studies show that irrespective of the micro geographic unit that has been examined, a very large proportion of crime is found at a small proportion of addresses, street segments, or clusters of street segments (Brantingham and Brantingham 1999; Pierce et al. 1988; Roncek 2000; Sherman et al. 1989; Weisburd and Green 1995). For example, Sherman et al. (1989) found that just 3.5 % of addresses produced 50 % of crime in Minneapolis during a single year (see also Pierce et al. 1988). Weisburd and Green (1995) found that approximately 20 % of all disorder crimes and 14 % of crimes against persons were concentrated in just 56 drug crime hot spots in Jersey City, New Jersey, an area that comprised only 4.4 % of street segments and intersections in the city. Eck et al. (2000) found that the most active 10 % of places (in terms of crime) in the Bronx and Baltimore accounted for approximately 32 % of a combination of robberies, assaults, burglaries, grand larcenies, and auto thefts. Similar findings have been reported about the city center of a large Metropolitan area in the United Kingdom: Bowers (2014) shows that from all facilities in which at least one theft occurred, about 80 % of all thefts occurred in about 20 % of facilities.

Looking at data across cities, Weisburd (2015) finds that there is a remarkable consistency, which he terms "a law of crime concentration" (see also Weisburd and Amram 2014; Weisburd et al. 2012). A total of 50 % of crime is found between 4 and 6 % of the street segments in the five larger cities examined using a consistent micro geographic unit and consistent definitions of crime. Between 2 and 4 % of streets generate 50 % of crime in three smaller cities studied. Weisburd (2015, see also Weisburd et al. 2004, 2012) also reports on the remarkable consistency of crime concentrations across *time* in four of the cities studied. Irrespective of fluctuations of rates of crime over time, the levels of crime concentration remain similar.

The importance of micro geographic units has been bolstered by strong evidence of the effectiveness of hot spots policing programs (Sherman and Weisburd 1995). In a systematic review of the evidence available, Braga et al. (2012) find that in 20 of 25 tests to date hot spot policing evaluations have yielded significant, positive evidence of deterrence. In all 10 randomized experiments, crime prevention gains were noted. Meta-analysis shows a consistent and significant impact of the approach on crime. Importantly, Braga et al. (2012; see also Weisburd et al. 2006) also find little evidence of displacement, and the meta-analysis reveals a statistically significant diffusion of benefits (see Clarke and Weisburd 1994) to areas nearby.

These basic and evaluation research findings have led a number of scholars to argue that much more attention needs to be paid to crime at micro units of geography. Nonetheless, to date only a very small proportion of the research literature in criminology is focused on

such units of analysis (Eck and Eck 2012; Weisburd 2015). Existing studies on micro places, as we detail below, provide strong support for the idea that focusing only on neighborhood or large area variability in crime misses a good deal of the variability of the crime problem. But to date scholars have not defined the extent of that loss, or the relative contributions to the crime problem at different units of geography, which may lead to misleading inferences of how place and crime interact. One reason for this gap in knowledge is that it is often not possible to compare geographic units. In the US, for example, where much of this work has been conducted, the commonly used unit for neighborhood or community boundaries is the census block group or tract. But micro geographic units such as street segments will often be inconsistent with these units (e.g. a street segment may be placed in two block groups or tracts).

While prior studies suggest the importance of examining crime and crime trends at a micro geographic unit of analysis, they have not compared explicitly the variability of crime per spatial level. If crime is highly concentrated within a small number of streets, but these streets in turn are concentrated within a small number of neighborhoods, then this favors neighborhood-level explanations of crime rather than explanations at smaller units. An important question is which part of the total variability of crime can be attributed to each spatial level, and to what extent such decomposition of variability has changed over time.

The Importance of Examining Micro Areas of Geography

The importance of studying crime at small units of geography has been recognized since the nineteenth century. However, the great majority of scholars with interests in geographic criminology were concerned with how crime varies across macro geographic areas such as regions, cities or neighborhoods. [For a comprehensive discussion of this subject for the criminology of place, we refer the reader to Weisburd et al. (2008) and Tita and Radil (2010), or for the 'modifiable areal unit problem' more generally, Openshaw (1984)] Glyde (1856) was the first scholar we could identify to question the validity of research findings when large areas were chosen as units of analysis in geographic criminology. In his paper, "Localities of Crime in Suffolk," he showed very clearly that larger units of analysis hide underlying variations in crime. Similarly, Henry Mayhew tried to uncover patterns in the distribution of crime in the city of London by combining ethnographic methods with statistical data. He interviewed prostitutes, criminals, and other citizens about alcoholism, poverty, housing conditions, and economic uncertainty. He focused on small areas like squares, streets, and buildings as units of analysis predating modern interests in the criminology of place by over a century.

Clifford Shaw was the first American scholar to recognize the importance of micro geographic units in the study of crime. Working as part of the Chicago School of sociologists who focused their interests on the importance of communities in the production of crime (Vold et al. 2002), he examined the home address of thousands of juvenile offenders on a map of Chicago. These were in many ways a precursor of the crime maps produced by recent scholars concerned with the criminology of place. While Shaw focused his theoretical interests on neighborhoods, the visual presentation data allowed the examination of variability at a micro geographic level.

Sustained interest in the geographic variability of crime at micro units of geography only began in the 1990s (Spelman 1995; Weisburd and Green 1995; Weisburd et al. 1992).

However, specific analyses of such variability have only been carried out recently. In a study looking at longitudinal crime trends at street segments in Seattle, Groff et al. (2009) suggest that biases in our understanding of crime are likely to result when the units of geography used for study are measured at such levels as census tracts or census block groups. They examine street-to-street variability in juvenile crime patterns across time in the city of Seattle over a 14-year period. Though they could not compare the contribution to the crime problem of street segments, their primary measure, with tracts or block groups because census units cut street segments into different units (only one side of a street segment is included in a census block) they use spatial autocorrelation approaches to examine the degree of clustering within areas and contrast that with the degree of independence of spatial patterns in crime at the micro geographic level. While they find that there is greater clustering of street segments with similar patterns or trajectories than would be expected by chance, their analyses show that there is also very strong street-to-street variability suggesting independence of street blocks in terms of crime patterns over time. Formal comparisons of trajectory groups reveal that they are independent of one another. Thus, it is likely that the specific spatial processes underlying the temporal patterns captured in trajectory analysis are slightly different for each of the trajectory groups. The finding of block-by-block variation in juvenile crime lends support for the examination of micro-level places. In sum, whatever the macro-level effects that influence crime across geography, these findings suggest that there are strong local-level trends that should not be ignored by researchers.

Weisburd et al. (2012) examined more general crime trends in Seattle over a 16-year period. Using Anselin's local indicators of spatial autocorrelation (LISA) (Bailey and Gatrell 1995; Rowlingson and Diggle 1993), they also examine whether the members of one trajectory pattern of crime over time are found near one another. Their findings reinforce the position of strong spatial heterogeneity at the street segment level. Street segments of one pattern are very likely to be surrounded by street segments of a different crime trajectory pattern. While there is evidence of clustering, or positive autocorrelation of trajectory patterns especially for the crime free pattern, their analyses overall suggest that crime patterns are interspersed throughout the city. Reinforcing this view is a cross-K-function analysis that allows examination of the type of relationships found in the 8 crime patterns examined in the study. None of the patterns evidence significant spatial repulsion at any of the distances examined. This suggests that whatever the strength of the influences on crime that are brought by larger geographic units such as communities or neighborhoods, local influences are producing strong variability of crime patterns at the street segment level.

Andresen and Malleson (2011) use a spatial point pattern test to investigate the (in)stability in the spatial distribution of crime over time at different levels of analysis in the city of Vancouver, British Columbia, Canada. Although they present some descriptive statistics on the percentage of street segments accounting for 50 % of crime—for example, in 1991 50 % of all burglaries were concentrated in almost 8 % of all street segments—their analyses focus on the change in spatial pattern over time rather than the variability of crime across spatial levels: they do not report the crime concentration for census tracts or dissemination areas. Nonetheless, they conclude that the street segment is the "driving force behind broader neighborhood change" (pg. 74), as they find more stable crime patterns across time for street segments than for larger spatial levels. Importantly, in Andresen and Malleson's (2011) analyses, they could not directly examine the nesting of smaller geographic units in larger units for the same problems of overlap that existed in the Groff et al. (2009) and Weisburd et al. (2012) analyses. Rather, their conclusions are drawn

from simply comparing the clustering and stability of clustering over time for each of the units separately.

Present Study

Groff et al. (2009), Andresen and Malleson (2011), and Weisburd et al. (2012) do not directly compare the relevance of micro versus larger geographic trends in a single analysis. A major reason for this is that standard definitions of larger geographic units lead to unit-independence problems. For example, the census area definitions are based on census blocks, which include the four block faces on a block unit. However, the street segment includes both block faces on two block units. The census basic units have been criticized by a number of scholars examining micro geographic units (Rengert et al. 2000; Weisburd et al. 2008, 2014). But irrespective of the value of the census divisions, street segments do not cluster directly within census block groups or tracts.[1]

The present study overcomes the problems of previous studies by using a geocoded crime dataset of more than 15,000 street segments, 114 neighborhoods, and 44 districts in the city of The Hague, The Netherlands. Street segments in these data are assigned to either one or another neighborhood (i.e. the street itself is not used for neighborhood delineation), and the city districts are perfect combinations of smaller neighborhoods. A multilevel dataset (street segments nested within neighborhoods nested within districts) can be constructed, which allows for statistical methods to decompose the total crime variance into the variance components per spatial level. Thus, we have the opportunity to examine variability at micro, meso and macro geographic units without a problem of overlap.[2]

Data and Methods

Crime Data

The present study uses a geocoded dataset (XY-coordinates) of reported crime in the city of The Hague, The Netherlands, spanning every year from 2001 to 2009. With a population of about 440,000 in 2001 to about 480,000 in 2009, The Hague is the third-largest city in the Netherlands (after Amsterdam and Rotterdam) and the seat of government in The Netherlands (CBS 2015).

For every year between 2001 and 2009, geocoded recorded crime data were provided by the unit of The Hague of the National Police, the Netherlands. The data refer to reported crimes by the public as well as crimes discovered due to police investigations, or crimes later admitted to by offenders themselves. Each crime event is geocoded to the XY-location where it occurred. As the purpose of the present study is to show how much of the total crime variability can be attributed to each level of spatial aggregation instead of the

[1] Andresen and Malleson (2011; p. 62) do not report how a street segment is attributed to census tracts: "Our criteria for a successful geocoded event is for the event to be geocoded to the correct 100-block (street segment). All data are geocoded to the street network and then aggregated to the census tracts and dissemination areas using a spatial join function." However, problems could arise if its two block faces fall within two separate census tracts, i.e. when a street is itself the border of a census tract.

[2] One caveat concerns where street segments *cross* neighborhood borders. We discuss the construction of the multilevel dataset in more detail in section "Units of Analysis".

crime variability across crime types, we combined crime types into one aggregate crime measure. Looking at a general crime measure also is required because specific crimes may be too rare at the micro geographies included in our analysis. Specifically, of the 162 crime types defined in the original dataset, we combined all crime types *except* for crime types referring to counterfeiting, fraud, driving while impaired by alcohol or other drugs, and environmental crimes because we thought these were not generally linked to specific geographic units. Thus, our aggregate crime measure refers to a sum score of 126 crime types, encompassing both violent crime and property crime. For 2001 to 2009, the total crime events are 48,940; 52,700; 54,954; 48,308; 40,572; 38,268; 40,782; 41,014; and 41,118 for a total of 406,683 crime events across the entire study period.[3]

Naturally, the police cannot always pinpoint the exact location of a crime event. Of the more than 400,000 crime events, 67 % could be geocoded to an exact location and about 1.5 % of crimes could not be geocoded at all (because there was no information about the location whatsoever). About 31 % of crime events could not be geocoded directly to a specific XY-coordinate, because only the street segment or named street (a combination of several street segments which bear the same name) was known. For crime events in which only the street name was known, the police used an approximation method: crime events are evenly distributed across a named street. For streets consisting of more than one street segment, this approximation may therefore result in an underestimation of the true number of crime events in one particular segment, while inflating the number of crime events in its adjacent street segment. On the other hand, this approximation method has the advantage that crime events are not artificially relocated to intersections, as is often the case in official US data.[4] It is important to note that because these crime events are distributed across street segments, the crime concentration at the street-segment level may be an *under* estimation of the true concentration. We investigate this further in sensitivity analyses of our main findings reported in section "Sensitivity Analyses".

Units of Analysis

We use three levels of spatial aggregation: districts, neighborhoods, and street segments.[5] Importantly as noted earlier, this is the first study we are aware of that is able to nest the three levels with relatively few problems of boundary overlap (see later). The city of The Hague is composed of 44 districts and 114 neighborhoods (administrative boundaries defined by Statistics Netherlands based on within-area homogeneity and following natural barriers). Although the 44 districts are sometimes further combined into 9 larger areas, official statistics are provided only for the district- and the neighborhood level. While the city of The Hague encompasses a total area of almost 33 square miles, the districts' area ranges from between 0.13 and 1.95 square miles (mean = 0.74 sq. miles), and the

[3] As in prior studies, property crimes including thefts make up a large proportion of the crimes studied. About 65 percent of the crimes examined here are non-violent thefts. Weisburd et al. (2004) found that about 50 percent of the crimes examined in their longitudinal sample in Seattle were non-violent property crimes.

[4] Visual inspection shows that the pattern of these approximated crime events is very similar to the pattern of non-approximated locations, leading to the tentative conclusion that there is no spatial bias in the location of approximated crime event locations. Some disparities also occur, especially near the coast.

[5] The street segment shapefile (Nationaal Wegenbestand of the year 2010) was obtained from the public domain under license http://creativecommons.org/publicdomain/zero/1.0/ from the Dutch National Geo-register, which facilitates access to publicly available datasets. The maintenance of the street segment shapefile falls under the Ministry of Infrastructure and the Environment. The neighborhood and districts shapefiles are obtained from Statistic Netherlands.

neighborhoods' area ranges from as small as 0.04 sq. miles to 1.30 sq. miles with a mean of 0.29 sq. miles. The average number of residents equals about 11,000 per district and 4400 per neighborhood.

In contrast to many US cities, The Hague does not have a grid layout, and city blocks are not used as unit of analysis in The Netherlands. Although this hampers direct comparison to international studies, this is an advantage for the purposes of the present study. The geocoded crime data do not refer to a block (with four block faces on one unit and a street segment encompassing both block faces of two adjacent block units). Instead, crimes are geocoded directly to an XY coordinate, which are located on street segments. There are 14,375 street segments in The Hague, with a mean length of about 94 meters (310 feet, with SD of 108 meters or 353 feet).

The Dutch definitions of the administrative neighborhoods either include or exclude an entire street (segment): street segments don't have one street face in one neighborhood and its other street face in another neighborhood. However, street segments do sometimes *cross* neighborhood borders.[6] In this case, we have split up the street segment into two separate street segments, one for each neighborhood, thereby allowing the creation of a hierarchical dataset of street segments nested in neighborhoods nested in districts. The XY coordinate of each crime event was then assigned to its new street segment. This procedure resulted in a final dataset of 406,683 crime events distributed across 15,527 street segments (M = 86 meters, SD = 96 meters) nested within 114 neighborhoods nested within 44 districts. Figure 1 shows the street segments (thin solid lines) nested in neighborhoods (short dashed borders), which in turn are nested in districts (thick solid lines).

Analytical Strategy

The analyses proceed in two steps. In the first step, we focus on describing the concentration of crime for different levels of spatial aggregation—districts, neighborhoods, and street segments—and their pattern over time. To this end, we provide standard summary statistics of the crime concentration but we also plot the cumulative percentage of spatial units against the cumulative percentage of crime, or Lorenz curve (Lorenz 1905). In addition to the ease of interpretation of the crime distribution across spatial levels of aggregation, the information in the Lorenz curve can be summarized by the Gini coefficient of inequality. The Gini coefficient is the ratio of the area between the line of perfect equality and the observed Lorenz curve to the area between the line of perfect equality and the line of perfect inequality (Gastwirth 1972). The coefficient thus varies between 0 (a completely equal distribution) and 1 (a completely unequal distribution with all crime concentrated in one unit). Gini coefficients are useful to discern trends in the crime concentration distribution over time (e.g. see Johnson 2010).

In the second step of analysis, we aim to quantify the amount of variability of the crime problem that is found across different units by employing hierarchical linear modeling [or linear mixed models, LMM, see, e.g. Taylor (2010) for a detailed discussion of multilevel models and their use for theory testing and development]. As the data refer to measurement over time per street segment, within neighborhood, within district, our data lends itself to panel data analyses. Our longitudinal analyses thus entail estimating a four-level model of years nested in street segments nested in neighborhoods nested in districts, to which a fixed effect of time can be added to capture an overall time trend (e.g. a decline in the number of crime events over the nine-year period). Allowing the slope of time to randomly vary

[6] This happens for 1152 of the 14,375 street segments.

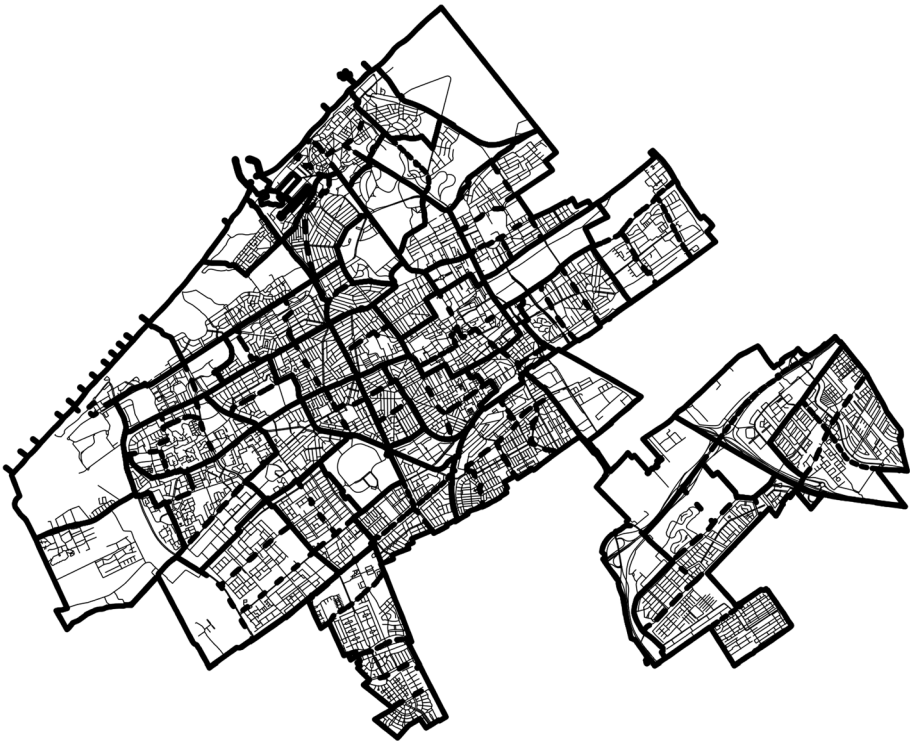

Fig. 1 Spatial units used in the study: street segments (*thin solid lines*) nested in neighborhoods (*short dashed thick borders*) nested in districts (*solid thick borders*)

across street segments (and/or neighborhoods and/or districts) captures different patterns of change over time per street segment (or neighborhood or district).[7]

The hierarchical model was estimated on a *sample* of our data for various reasons. First, an assumption of the random effects model is that the units for each level of data are a random sample of the population rather than the entire population. We therefore preferred to model a sample of our data rather than assume that our population data are actually a sample of a hypothetical larger population. Second, raw processing power was a concern. Estimating a 4-level model with random slopes for time for 9 time periods within 15,527 street segments, 114 neighborhoods and 44 districts takes a very long time, as did the exact likelihood ratio tests (see below) using bootstrapping. Third, as crime is concentrated on a very local spatial level, including the entire population of street segments would lead to spatially correlated errors on the street segment level. Not accounting for these potentially correlated random effects could lead to incorrect inferences on the variance proportion attributed to each spatial level. By drawing a sample of street segments, we largely counter such concerns.

[7] Of course several studies have modeled crime using HLM, e.g. street segments nested in output areas, nested in medium super output areas (Davies and Johnson 2015). However, the focus of these studies is often on obtaining parameter estimates of covariates and not on the variance decomposition of crime per spatial level and its change over time.

Analyzing a sample of data can also have disadvantages, in particular because the sample may not be valid representation of the full dataset. Because the estimated model is tailored to the specific sample, inferences may be incorrect. Therefore we used a bootstrapping procedure: 500 stratified random samples of 25 % of street segments per neighborhood were taken and each of the samples was analyzed (each sample consisted of 9 time periods within 3857 street segments, 114 neighborhoods, and 44 districts). The reported variances and variance proportions in the results section are thus the mean across 500 separate model estimations. "Appendix" shows that for all variance estimates, the number of replications does not affect results greatly, and that 500 replications are thus more than adequate to attain stable results.

Estimation proceeded in several steps. First, we investigated whether the distribution of the dependent variable's residuals violated the assumptions of the linear model. For each wave of data separately, we estimated a three-level multilevel model (street segments nested in neighborhoods nested in districts) for the raw crime counts, and logged crime counts. Residual diagnostics plots using the raw crime count revealed severe violations of normality assumptions, while model violations are mild for the logged crime measure, allowing us to proceed with LMM and log(crime + 1) as the dependent variable.[8]

We used Likelihood Ratio Tests (Chi square distribution test) to investigate whether decomposing the panel data into two, three, and four levels improved model fit, the standard approach in most random effects literature. Following concerns about the standard LRT (e.g. see Long 2011, Section 10.4), we also use (exact) restricted likelihood ratio tests based on simulated values from the finite sample distribution, specifically using the *exactRLRT* function from the RLRsim package (version 3.1-2) in the R statistical environment (version 3.2.2). The complete model, using 4 levels, fit the data best.

As we are interested in capturing the variability of crime rather than explaining this variability by time-constant or time-varying covariates, we do not add predictor variables—with the exception of the length of each street segment, as longer street segments naturally provide more opportunities for crime to occur than shorter street segments. Descriptive plots suggested a logged version of road length, while model comparisons suggested a cubic effect.

Finally, we investigated the effect of a time trend. Model fit comparisons suggested that a linear effect adequately captured the trends of crime over time. The coefficients of time were allowed to randomly vary across street segments, neighborhoods, and districts, modeling diverging time effects for each spatial level. The final model has t measurements nested within street segment i nested in neighborhood j nested in district k,

[8] An alternative approach of the log-transformation is to use generalized linear mixed models (GLMM), which directly model the non-normal error distribution of the response variable. For our (count) data, one possible GLMM uses a log link function and the probability mass function for the Poisson distribution. A disadvantage of GLMM is that inference is much more complicated than for linear models. The likelihood can only be approximated, e.g. using numerical integration, in contrast to LMM. While the accuracy increases as the number of integration points increases, the number of evaluations also increases exponentially with the number of random effects; the size of the dataset also impacts computation time a lot. For this reason, the statistical package used in this paper (lme4 in R) only allows a single integration point (i.e. Laplace approximation) when estimating more than one random effect. Moreover, it is assumed that the sampling distributions of the parameters are multivariate normal, and this is unlikely with fewer units per level, although confidence intervals for random effects can be approximated using (e.g.) parametric bootstrap methods (Efron 1979). Finally, a disadvantage of GLMM is that, in contrast to LMM, the level-1 variance depends on the expected value and is therefore not reported by most statistical software—although a simulation method has been proposed as a solution (Browne et al. 2005). In short, residual diagnostics plots indicated that a linear mixed model can adequately model a logged version of crime, and we therefore present these results instead of GLMM outcomes.

$$\begin{aligned}\log(Y_{tijk}+1) &= \beta_{0tijk} + \beta_{1ijk}time_{tijk} + \beta_2 \log(length+1)_{ijk} \\ &\quad + \beta_3 \log(length+1)^2_{ijk} + \beta_4 \log(length+1)^3_{ijk} \\ \beta_{0tijk} &= \beta_0 + f_{0k} + v_{0jk} + u_{0ijk} + \varepsilon_{0tijk} \\ \beta_{1ijk} &= \beta_1 + f_{1k} + v_{1jk} + u_{1ijk},\end{aligned}$$

with correlated random effects, estimated using restricted maximum likelihood using the *lmer* function in the lme4 package (version 1.1-10) of R (version 3.2.2).

For the purposes of this study, we are only interested in the variance of the random effects. The estimated district-level variance σ^2_{f0}, the neighborhood-level variance σ^2_{v0}, and the street segment-level variance σ^2_{u0} allow inferences of the crime variability, because they show the proportion of the total variance in crime that can be attributed to each spatial level. Importantly, the random slopes of time imply that these variances are not constant but depend on time. We will therefore present the variance function of each spatial level as it changes over time.

Results

Before we turn to descriptive statistics of the data, we first visualize the crime counts on the map of The Hague. Figure 2 shows the number of crimes per square mile for each of the 44 districts in 2001, 2005, and 2009. The legend is based on the Fisher–Jenks algorithm (Fisher 1958; Slocum et al. 2005), which has classified the districts into eight classes such that the sum of squared deviations from the class means is minimal. Clearly, some districts are safer than others. Figure 2 also shows that over time, crime has become less concentrated in some districts, showing a more equal distribution of crime counts across districts in 2009 than in 2001. Figure 3 shows the number of crimes per square mile for each of the 114 neighborhoods in The Hague. On this spatial level of aggregation, not much change in the spatial pattern of crime events can be detected by the naked eye. Compared to Fig. 2, these visualizations suggest that changes in the crime concentration are more strongly divided between-districts than within-districts, between neighborhoods.

We also plotted descriptive maps for the street segments, but these are difficult to read and therefore not printed here. The Hague follows descriptions of street-to-street variability of crime in the US (e.g. see Weisburd et al. 2012) and the U.K. (e.g. Johnson 2010). Crime varies a good deal across street segments in The Hague with some streets segments never experiencing any crime over a nine-year period, whereas other street segments experience many crimes. The difference between street segments is very large: on at least one street segment, 4371 separate crime events occur over the nine year period and on many streets there was little or no crime. Similar to Johnson (2010), these results show that "the explanation for the distribution of crime cannot be found entirely in theories which consider sociological processes that operate at the area level." There are clearly local as well as larger area effects at play. The quantification of these effects are the subject of the remainder of this study.

Descriptive Statistics

We first present descriptive statistics on the percentage of spatial units that accounts for 50 % of all crime. For brevity, we only present results for 2001, 2005, and 2009. In all

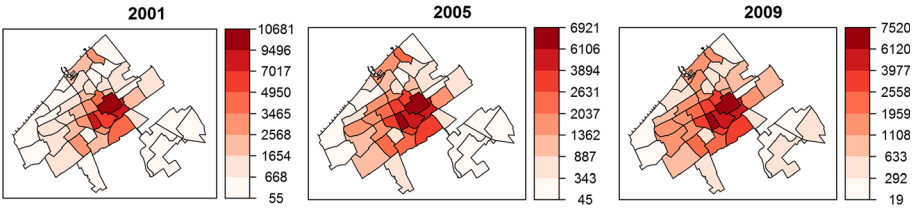

Fig. 2 Crime rates (count per sq.mile) in The Hague, The Netherlands per district (2001, 2005, 2009). *Note* Intervals are based on the Fisher–Jenks algorithm (Slocum et al. 2005)

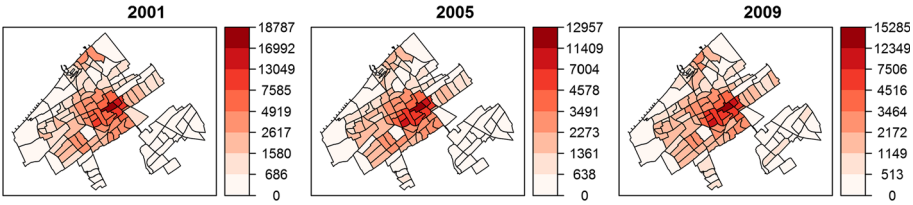

Fig. 3 Crime rates (count per sq.mile) in The Hague, The Netherlands per neighborhood (2001, 2005, 2009). *Note* Intervals are based on the Fisher–Jenks algorithm (Slocum et al. 2005)

waves of data, a very small percentage of street segments account for the bulk of the crime, and these distributions follow the *law of crime concentration* described by Weisburd (see Weisburd 2015; see also Weisburd et al. 2012; Weisburd and Amram 2014). Column (a) of Table 1 shows that 50 % of all crime occurs in about 6.3–7.3 % of all street segments, or about 1000 of the 15,527 segments. In contrast, about 17.5–20.2 % of neighborhoods (n = 20–23 of N = 114) or 15.9–20.5 % of districts (N = 7–9 of N = 44) are responsible for 50 % of all crime.

Following Andresen and Malleson (2011), in column (b) of Table 1 we also report the percentage of spatial units that have at least one crime occurrence. For street segments, almost one half of The Hague is free from crime across the study period. In contrast, almost every neighborhood (112 in 2001 and 113 in 2005 and 2009) experiences at least one crime, and every district has at least one crime occurrence. These results are remarkably in line with results from Vancouver (Andresen and Malleson 2011).

Table 1 Percent of spatial units accounting for 50 % of crime

	(a) Percentage of spatial units accounting for 50 % of all crime			(b) Percentage of spatial units that have any crime			(c) Percentage of spatial units with crime that account for 50 % of all crime		
	2001	2005	2009	2001	2005	2009	2001	2005	2009
Street segment	6.27	7.26	6.83	52.32	52.54	52.28	11.98	13.83	13.07
Neighborhood	17.54	20.18	20.18	98.25	99.12	99.12	17.86	20.35	20.35
District	15.91	18.18	20.45	100	100	100	15.91	18.18	20.45

N = 15,527 street segments, 114 neighborhoods, and 44 districts

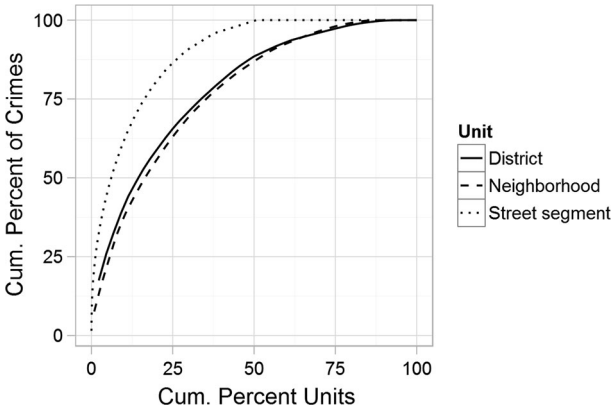

Fig. 4 Lorenz curves for 2001 crime data

Finally, column (c) in Table 1 shows, of those non-zero spatial units, the percentage of units that accounts for 50 % of all crime. As almost all neighborhoods and all districts experience at least one crime event in each wave of data, these percentages do not differ much from column (a). For street segments, e.g. in 2001, we see that within the 52.3 % of street segments that experience at least one crime event (or N = 8124 street segments) about 12.0 % (or n = 973) account for 50 % of all crime. Thus, even when focusing only on places that have any crime, there is a high(er) degree of crime concentration at the street segment level (than at the neighborhood or district level).

We next present the entire crime concentration distribution in Fig. 4. The advantage of this presentation is that no arbitrary decision has to be made to report the percentage of spatial units that account for 50 % of all crime, or the percentage of crime that is accounted for by 20 % of all spatial units (see also Johnson and Bowers 2010; Davies and Johnson 2015). Points on the Lorenz curve allow statements like "25 % of the units of analysis are responsible for 50 % of all crime" (green line), or "100 % of all crime is concentrated in 60 % of all units" (red line). An important takeaway from Fig. 4 is that crime is concentrated on every level of spatial aggregation, and crime is even more concentrated on the street-segment level than on the higher-spatial levels. Our findings are similar to earlier studies in the UK. For example, Johnson (2010) examined patterns of crime concentration at the street segment and census area levels using these approaches, and showed that burglary is more concentrated at the street segment level than at the census area level.

We next turn to the development of the degree of crime concentration across different spatial levels over time. Although we could overlay Lorenz curves of each wave of data, the resulting figure is difficult to interpret. We therefore summarize the crime distributions in the Lorenz curves using the Gini coefficient (with 0 indicating a completely equal distribution of crime events across units of analysis, and 1 indicating that all crime occurs in just one unit of analysis). The Gini coefficient is the ratio of the area that lies between the line of equality (i.e. the diagonal in Fig. 4) and the Lorenz curve over the total area under the line of equality.[9] Figure 5 shows that crime is highly concentrated and rather

[9] Johnson (2010) points out that the Gini coefficient is dependent upon the shape of the estimated line of equality, the shape of which varies for different units of analysis. More specifically, Johnson (2010) encounters this problem because he studies home burglary and the number of homes is unequally concentrated per street segment. Here, we only use the Gini coefficient as a tentative measure of crime concentration across street segments, and use HLM to address the same question more directly.

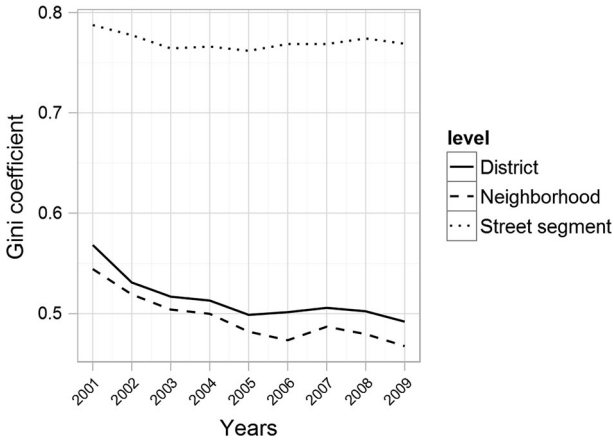

Fig. 5 Gini coefficients for all crime (2001–2009), for all spatial units

stable over time for street segments. The concentration of crime for neighborhoods and districts is still quite substantial (around the 0.5 mark on a scale of 0–1), although crime is noticeably less concentrated on these spatial levels. In addition, there seem to be a slight downward trend over time for these units, indicating that over time crime becomes somewhat less concentrated within neighborhoods and districts.

Another preliminary conclusion from Fig. 5 is that the degree of crime concentration at the neighborhood-level does not differ much from the district-level. This may indicate that there is relatively high within-district (between-neighborhoods) homogeneity in crime concentration. Thus, the neighborhood-level may not account for much above and beyond the crime concentration at the district level.

Hierarchical Linear Models

Although the descriptive results already provide an indication about the amount of crime variability per spatial level, these statistics do not capture this amount directly. The disadvantage of the previous methods is that analyses are performed separately for each spatial level (disregarding the nested structure of the spatial units) as well as separately for each wave of data. Hierarchical linear models provide a convenient way to assess the proportion of the total crime variance that is accounted for by each level of analysis.

As discussed in the *Analytical Strategy*, we estimated a four-level model of waves of data nested within street segments, nested within neighborhoods, nested within districts. The variances for each (spatial) level then allow for inferences of the crime variability, because they show the proportion of the total variance in crime that can be attributed to each spatial level. Because of the random slopes of time on each spatial level, the variance on each level is non-constant, but dependent on time. For example, the variance for the street level is estimated by:

$$var\left(u_{0ijk} + u_{1ijk}time_{tijk}\right) = \sigma_{u0}^2 + 2\sigma_{u01}time_{tijk} + \sigma_{u1}^2 time_{tijk}^2$$

with σ_{u01} referring to the covariance between the random effect of street segment and the random slope of time. The other variance functions are estimated equivalently.

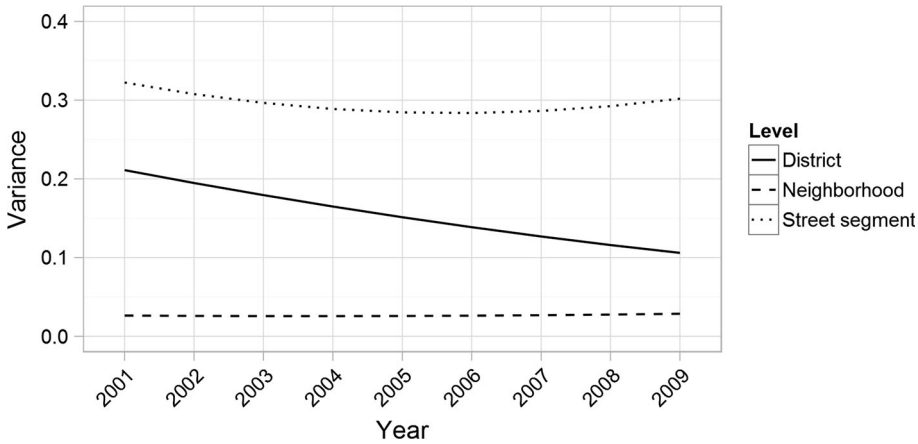

Fig. 6 Variance functions per spatial level (2001–2009)

Figure 6 shows the variance functions at the three spatial levels as a function of time. The neighborhood and district variances, while statistically significant, do not show substantive non-linearity over time. Moreover, the variance that can be attributed to the (meso-) neighborhood level, is consistently very small over time. The variance of crime at the district level on the other hand *decreases* substantially over the nine-year period, indicating increasing within-district variability in crime over time. The street segment level is clearly where most of 'the action' is in crime. Over time, the model suggests a non-linear change in the variance, with first decreasing and then increasing variance on the street segment level.

To ease interpretation, Fig. 7 shows the variance converted into the proportion of the total variance that can be attributed to each variance component.[10] On average across all years, about 62 % of the total variance can be attributed to variation between street segments. Districts account for about 32 % on average, while between-neighborhood variance (controlled for street segment and district-level variability) only accounts for about 6 % of the total variance. The results indicate an increase in the variance proportion—or, the degree of crime concentration—on the street segment level over time: from 58 % in 2001 to 69 % in 2009. The proportion of total variance accounted for by the neighborhood-level remains rather stable between 5 and 7 %, while the district-level variance proportion decreases from 38 % in 2001 to 24 % in 2009.

The results confirm the descriptive statistics in that crime is most heavily concentrated on the street segment level, with also a substantive amount of concentration on the district level, whereas crime is distributed more equally across neighborhoods. Substantively, this means that some districts as a whole are more crime-prone than others, but that neighborhoods within the same district are rather homogeneous with regard to crime (either high or low). By far most of the action occurs on street segments, and the importance of street segments increases over time.

[10] There is also a residual variance component of time, capturing the variance of crime that can be attributed to time-varying explanations. The proportions reported are the proportion of variance in crime of each spatial unit as compared to the total variance explained by the three spatial units.

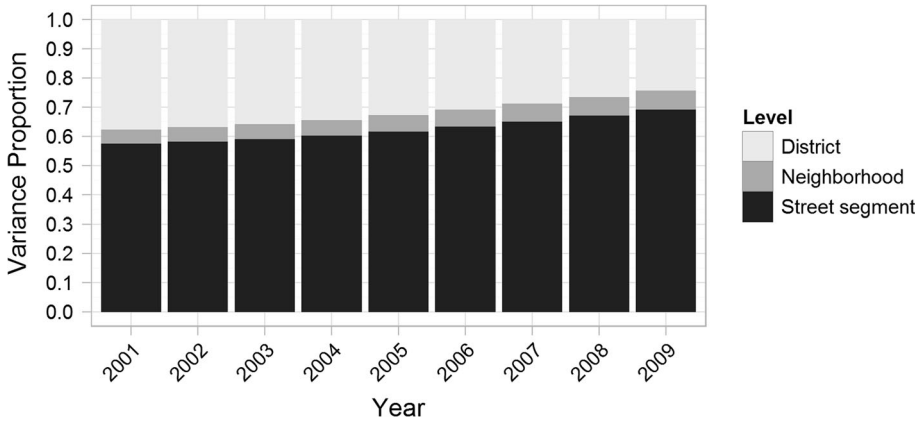

Fig. 7 The proportion of total variance attributed to spatial levels (2001–2009)

Sensitivity Analyses

To assess the robustness of our outcomes as well as to test the implications of several research decisions, we conducted three types of sensitivity analyses. First, we investigated the robustness of outcomes against outliers. The distribution of crime shows some outliers; for example, in 2001, one street segment experienced 722 crime events. In additional analyses, removing these street segments did not affect our results. As an alternative approach, we truncated street segments with such high crime counts to 200 crime events, and this decision also did not affect the above results. Similarly, a few street segments in the dataset are actually highway access points (identified in part by the long length of these segments). Removing these street segments from analyses also did not affect substantive results.

Our second sensitivity analyses concern the procedure employed by the police to distribute crimes evenly on the 'named street' (which may consist of more than one street segment) if the location could not be assigned to a nearby property address. Of course for named streets that consist of just one street segment, the crimes allocated to that street based only the street name can be considered 'accurately' geocoded for the purposes of this study. For all other crimes, we argued that because of the location approximation procedure, "the crime concentration at the street-segment level may be an *under*estimation of the true concentration". In additional analyses, we tested this argument directly. In line with our expectations, our new analyses show that for the "accurate" geocoded crimes, the proportion of variance at the street segment level is larger than our original analyses. The proportion of variance per level of analysis varied between 2001 and 2009 from about 71–77 %, 4–6 %, and 25–17 % for street segment, neighborhood, and district, respectively. In short, these analyses show that when analyzing this subset of crimes, street segments are even more important.

The third additional set of analyses concern our choice to analyze crime as a single measure rather than categories of crime. If different crimes cluster differently, and these crime clusters are located in different locations, then merging all crime types can create a situation where an aggregate area of clustering is actually comprised of more micro-level spatial clusters of different crime types. Of course, in that case the reported analyses would *under*estimate the importance of street segments: any inferences on the importance of micro geographic units vis-à-vis meso units and macro units will be conservative, as the

variability within-neighborhoods between-street segments is inflated to the extent that the different types of crime cluster at different micro-locations. To test this argument, we analyzed *property* crime, a combination of crime types that involve the taking of property without force or the threat of force (e.g. burglary, larceny, theft, and shoplifting). Results indicate that 'the action' in property crime is even more on the street segment level than the results reported for total crime. The proportion of the street segment-level variance as compared to the total variance that can be attributed to all spatial units is about 62 % in 2001, increasing to 74 % in 2009. Following the logic of the previous paragraph, we finally analyzed only those property crime events that had not been geocoded to an approximate location. Results show much higher variance proportions for street segments than in the previous analysis, ranging from about 73 to 80 % for street segments. In summary, additional analyses on a subset of crime events strengthens our conclusions about the importance of micro areas for the explanation of crime, and street segments matter even more for property crimes than for a combined measure of all crimes.

Discussion

With the growth of interest in micro geographic units over the last decades, scholars have begun to make the case that examination of crime at neighborhoods or larger macro geographic units may miss a good deal of the action of crime (Andresen and Malleson 2011; Groff et al. 2009; Weisburd et al. 2012). But analyses so far have not been able to compare the contribution of different units to the crime problem because of a lack of unit independence. Our study is the first we know of to directly examine this question, looking at micro geographic units nested within neighborhoods and larger macro geographies.

Our descriptive findings confirm prior studies that suggest that leaving out the micro geographic level will lead to a loss of important information about the crime problem. Crime is concentrated at every unit of analysis observed. Across streets, neighborhoods and districts crime is concentrated at a relatively small number of units. Nonetheless, even at this descriptive level the concentration is greatest at street segments.

Our findings examining the hierarchical structure of crime across geographies provides important new data because they suggest the relative degree to which each unit of geography contributes to the distribution of the crime problem. What is clear is that street segments overwhelm higher geographic units in their contribution to variation in crime across the city. Between 58 and 69 % of the variability in the crime problem is accounted for at the street segment level. The bulk of the remaining variance is contributed by the district level of analysis.

The salience of street segments in our analyses certainly calls for greater focus on study of micro geographic units in crime. Weisburd (2015) reports that fewer than 5 % of the empirical studies in *Criminology* examine crime at micro geographic levels. Eck and Eck (2012) find even fewer examples in journals examining crime policy of studies that consider micro geographic units. Clearly Weisburd's (2015) call for criminologists to make the scene of micro geographic study is reinforced by our work. It is time to identify why street segments produce such a large part of the variability of crime in cities. It may be that there are elements of explanation of this concentration that relate simply to land use or other structural characteristics of modern city planning. But certainly it draws our attention to social and opportunity characteristics of places (see Weisburd et al. 2012). With the growing interest in routine activities and crime (Cohen and Felson 1979) it is natural to bring one's focus to

micro geographies because that is where the specific opportunities for crime are located. Weisburd et al. (2012) suggest that there is also important variability in such social characteristics as poverty and collective efficacy at the street segment level. But whatever the suggestions of recent studies, we know too little about the "why" of crime concentrations at a micro geographic level. If this level is the primary contributor to the crime problem, then in understanding crime and crime trends it is time to focus more effort on this level of analysis.

One surprising finding from our work is that the neighborhood level as it is defined in The Hague does not add much to the action of crime beyond the larger district areas. We are hesitant in drawing too strong conclusions from these findings. Nonetheless, it does raise the question of whether larger geographic definitions of communities are better than smaller units that have become popular in criminological studies in recent years (e.g. see Clear et al. 2003; Gorman et al. 2001; Sampson and Raudenbush 2004). Census block groups are somewhat similar in size to our neighborhood areas. But our analyses suggest that that may be too large a geography to capture micro geographic trends and too small a geography to capture macro geographic trends. Perhaps criminologists should begin to examine more critically whether census block groups are an appropriate geographic level for understanding crime. Indeed some scholars in this area combine census units with the idea of creating boundaries of communities that are more consistent with the theoretical interests of researchers (e.g. see Sampson and Morenoff 1997).

We think our findings are very much in line with the observations of the Chicago School that brought interest to communities and crime in American sociology and criminology. Park (1925) defined the neighborhood as the elementary form of cohesion in urban life. But such neighborhoods were not of the meso geographic variety of recent studies of census block groups. Rather they encompassed broad areas of the city. Ernest Burgess, drawing from an inventory of price changes in housing values in Chicago areas developed a concentric zone model of the distribution of social problems and crime for cities (especially for Chicago) (Burgess 1925/1967).[11] Burgess suggested that Chicago included five concentric[12] zones, each containing various neighborhoods, four of them situated around 'The Loop' (the business center of the city): "the typical processes of the expansion of the city can best be illustrated, perhaps, by a series of concentric circles, which may be numbered to designate both the successive zones of urban extension and the types of areas differentiated in the process of expansion" (Burgess 1925/1967: 50). Importantly, Burgess' unit of analyses was a series of neighborhoods within cities that share similar characteristics. We think that this conception of large areas with similar social structures is closer to the macro geographies we identify as important in our work than the meso geographies of neighborhoods as defined in The Hague.

We recognize that the neighborhood and district definitions in The Hague may not capture the specific geographic units most important to crime. Perhaps if criminologists were defining these units, they would have used very different boundaries, recognizing that community is not just about homogeneity in specific census measures, but also about the dynamics of crime in the city (Sampson 2012). Accordingly, our data do not prove that the action of crime is primarily at the street segment as compared, for example, to the criminological conception of neighborhoods. But they do raise a challenge to criminologists to identify whether better definitions of community will move the action of crime more to the community level.

[11] The real estate agent had discovered zones in the city of Chicago when he made up an inventory of price changes of houses and real estate. He contacted Burgess regarding his findings, which led to the now famous geographic model of crime and social problems in the urban context.

[12] In reality only half circles because Chicago is situated at the border of Lake Michigan.

Our work also suggests that 2001–2009 experienced a crime trend towards stronger concentration at a local level like street segments. While we cannot identify directly from our data the cause of this increasing trend in concentration, we think it interesting to speculate on potential explanations. One explanation may be the crime trend itself. While our findings suggest that we need to turn to micro geographies to understand the variability of crime, and accordingly crime trends, it may be that the decreasing crime trend itself causes concentration. However, a decreasing crime trend by itself does not translate necessarily into increased variability at the lowest level of the hierarchical model. In addition, the importance of street segments for the explanation of crime increases fairly steadily over time, whereas the crime trend itself is not linear across the time series, not lending much weight to this explanation.

A closer look at the data suggests another explanation. The results are clear about where this trend towards increasing concentration at the street segment level comes from: from a decrease in the between-district differences (or: an increase in the within-district, between-street segment differences). What this suggests is that the districts became less homogeneous in terms of crime in this period. Although this finding for crime could reflect a trend toward greater within-district heterogeneity in terms of other social characteristics, we do not have empirical data at the street segment level to provide direct evidence of such changes.

While we think that our work advances knowledge of crime at place, we want to note before concluding some limitations of our work. First, much research on crime has been conducted in the US, and our results from The Hague may simply reflect the differing structures of American and European cities. Perhaps the dominance of micro geographic units in understanding the variability in crime would not be similar in the US. The importance of macro as opposed to meso distributions may also be different in US cities. We suspect not, given the growing body of evidence suggesting the importance of micro geographic trends, but nonetheless this work should be replicated in the US and elsewhere before drawing unambiguous conclusions.

Finally, while we think our study provides important new data to our understanding of crime at different levels of geography, we recognize that our decision to study crime as a general measure and as a category of 'property crime' limits to some degree our ability to draw inferences from our data. Our work does not refer to clustering of very specific types of crime, for example home burglaries or crimes of violence such as assaults. Perhaps the nature of the contribution of different levels of geography to the crime problem would be different for different types of crime, which could be investigated in future research. The methodological limitation in our study was that using specific measures of crime would have led to distributions that could not be easily examined (in terms of explained variance) within the current hierarchical analytic framework. Moreover, our study is consistent with many others that have begun to examine crime at a micro-level of geography (e.g. see Groff et al. 2009; Weisburd et al. 2012). The reality is that data of this type become sparse when specific types of crime are examined.

Conclusions

The study of crime has begun to include a focus on micro geographic units. At the same time, existing studies do not allow us to compare different units in terms of their contribution to the crime problem. Our examination of crime in The Hague allowed us to

examine the contribution of street segments, neighborhoods and districts to the overall distribution of crime in the city. Our findings strongly reinforce the importance of studying crime at micro-levels of geography. Indeed, our hierarchical analyses suggest that street segments provide by far the most significant contribution to the action of crime in the city. This does not mean that it is unimportant to examine crime and crime trends at macro-levels of geography. Indeed, our work suggests that an approach that examines crime at a micro-level of crime like street segments combined with a large area approach, such as that represented by districts in The Hague, may be most efficient for understanding and doing something about the crime problem. But, it is certainly time for criminologists and policy makers to take note of the importance of crime at a micro geographic level.

Appendix

See Fig. 8.

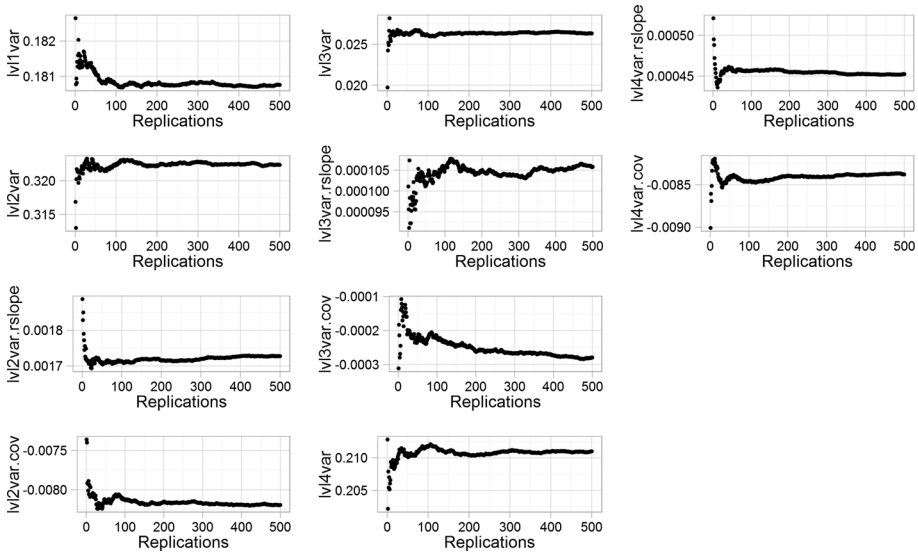

Fig. 8 Mean estimate of variance estimates as a function of number of replications (all crimes)

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